While you do have visibility into network interfaces (both configuration and dynamic information) in the case of a VM or a physical server, in the context of a cloud function or lambda function, you do not.

3.4.2 Higher-level network sources

In contrast to network interfaces, higher-level network sources, such as load balancers, are usually directly accessible in terms of having access to the relevant signals, most importantly metrics (health and performance related). Load balancers, both L3/4 and L7 (that is, HTTP level) typically provide a wealth of signals—for example, the Amazon Application Load Balancer (ALB; http://mng.bz/0KvJ) supports logs, metrics, and traces. For a concrete example, see figure 3.6, which is derived from an open source observability demo (https://github.com/observe-k8s/Observe-k8s-demo). The same is the case for the Amazon API Gateway (http://mng.bz/9Dpj) in front of lambda functions. One important aspect in this context is the health of the backends and how to make sure they are up—and, hence, traffic can be routed to them. For example, an `HTTPCode_ELB_504_Count` code from the ALB (http://mng.bz/Wz4g) indicates that backend resources didn't respond in time or, as shown in figure 3.6, various usage-related metrics. Further, you should consider that certain orchestrators, such as Kubernetes, introduce network abstractions (http://mng.bz/8reK)—for example, an Ingress resource—allowing you to manage higher-level network resources in clusters along with signals that may or may not be exposed with the abstractions.

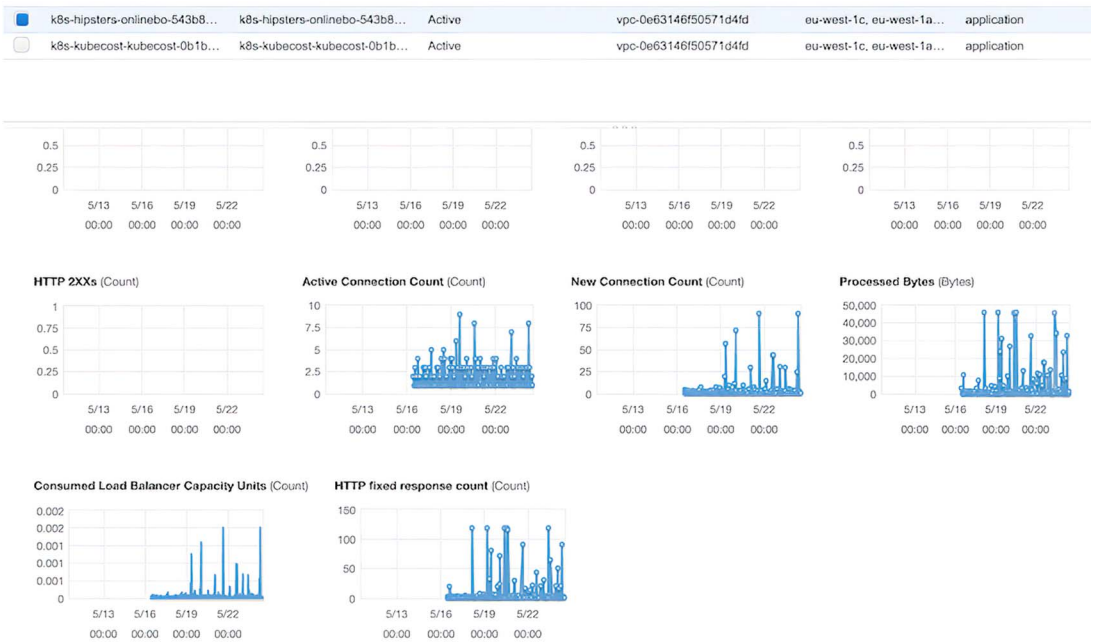


Figure 3.6 Screenshot of the ALB-monitoring console, showing usage metrics